Anthony Firn

Lead Software Engineer — full-stack, platform, AI and graphics

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Remote, US

SUMMARY

Lead engineer who designs, builds and runs production systems end to end: customer-facing web apps, the data pipelines behind them, and the cloud infrastructure they deploy to. Equally at home in Python, TypeScript and Terraform as in GPU simulation, WebAssembly and game engines. Known for shipping working software, documenting it, and keeping it running.

EXPERIENCE

Lead Engineer · True Native Media

2025 – Present

- Lead development of AdVero, the web workspace podcast shows, advertisers and staff use to run campaigns, approve ad reads and report issues.
- Built the campaign-planning data platform: Cloud Run jobs and services for inventory measurement, quotes and sync orchestration, with audits that record production data changes.
- Designed serverless Airtable, Firestore and BigQuery sync pipelines with schema-drift detection; manage Google Cloud infrastructure as code with Terraform.
- Stood up self-hosted tools on Google Cloud, including a support desk and per-user AI agent workspaces with approval controls, plus email automation with daily health reports.

Systems Engineer · MedSpark

San Luis Obispo, CA

- Worked across the whole product system for a medical product development firm: hardware design, firmware, software and server infrastructure.
- Ran precision manufacturing, including wire EDM machining, to turn designs into working parts.

SELECTED PROJECTS

all open source, live at evolvetech.dev

Retirement Research

Python · PyTorch · FastAPI

- GPU simulator comparing retirement withdrawal strategies by spendable income after tax across 2,000 reshuffled 40-year market histories and seven stress cases; reproducible pipeline and an in-browser explorer.

Emberwake

Godot 4 · Blender · WebAssembly

- 3D action roguelike with procedural rooms and audio, reproducible Blender-scripted assets, a TCP agent bridge for AI play-testing, and a strict-CSP web export.

Symmetry Forge

Python · Pyodide · Web Workers

- Toolkit for graph automorphisms and combinatorial designs; the unmodified package runs in the browser via Pyodide in a cancellable Web Worker.

evolvetech.dev hosting

GitHub Actions · Netlify · Bunny · AWS

- Dual-CDN static hosting with health-probed DNS failover, timed recovery, and DNS-validated certificates installed on both providers; verified with a live failover drill.

SKILLS

Languages	Python, TypeScript, JavaScript, SQL, GDScript, Bash, Verilog
Web & mobile	React, Next.js, React Native / Expo, FastAPI, Apps Script, Capacitor
Cloud & platform	Google Cloud (Cloud Run, Functions, Pub/Sub), AWS, Terraform, GitHub Actions, Netlify, Bunny CDN
Data	BigQuery, Firestore, Airtable, NumPy
AI / ML	PyTorch / CUDA, vLLM, LLM agents, MCP, Open WebUI
Graphics	Godot 4, Blender, WebGL / Canvas, WebAssembly, Pyodide

EDUCATION

B.S. Computer Engineering
California Polytechnic State University, San Luis Obispo